

Predictions of Small-Body Physical Traits Based on Orbital Characteristics

Micro-Project 1

[GitHub Repository](https://github.com/Ethereal-Mining-Company/small-body-analysis)

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Problem Statement

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Physical prospecting and exploration of small-bodies incur heavy operational and capital costs, making the early detection of non-viable targets critical. Small-body volumetric size is a primary variable for predicting economic viability. A small-body that fails to meet the minimum size threshold will not meet viability requirements even if the small-body has exceptional mineral composition. To reduce the risk of prospecting and exploring non-viable small-bodies, a data model must be created to filter out sub-threshold small-bodies.

Peer-reviewed research has determined certain orbital characteristics such as albedo, absolute magnitude (H), semi-major axis (a), eccentricity (e), and inclination (i) can be used to increase accuracy for estimating small-body size (Ge, 2026). These characteristics have been broken down into three broad categories based on the relative cost and reliability of observing these characteristics. Group 1 characteristics are low cost to observe and the observations have high reliability (Murcia, 2023). Group 2 characteristics are low cost to observe but have medium reliability because the observations are highly variable and prone to observational error when gathering (Robinson, 2024). Group 3 characteristics are high cost to observe and have high reliability (Ge, 2025).

The model must be cost-effective in determining small-body size accurately. Group 3 characteristics are unlikely to be cost-effective and there is limited access to the equipment because access is allocated based on competing use requests from other higher priority projects (NASA, 2025). Group 1 and Group 2 characteristics provide the best candidates for cost-effective observations, however, since Group 2 includes characteristics that are prone to observational error a cost-benefit analysis will determine if excluding these observations will decrease accuracy in determining the size of a small-body beyond the statistical and operational acceptable thresholds.

A two-model comparison is necessary to determine what model will create the most cost-effective system for filtering out sub-threshold small-bodies. The first model will use regression analysis via machine learning with Group 1 and Group 2 characteristics to predict the diameter of a small-body. The second model will act as a scientific baseline, using regression analysis via machine learning with Group 1 characteristics only to predict the diameter of a small-body. Because current peer-reviewed research suggests absolute magnitude (H) (Group 2 characteristic) is mathematically required to calculate diameter (Ge, 2026), this analysis will determine if relying solely on highly reliable Group 1 characteristics results in a failure to accurately predict diameter. Each model will be assessed for overall accuracy to predict the size of a small body. An analysis of the accuracy to predict the diameter of small-body will determine if the operational trade-off between feature measurement cost and predictive performance of using Group 2 characteristics will have a material impact on properly filtering sub-threshold small-bodies.

Once the two-model comparison is complete the results will be used to create the final model using the more cost-effective characteristics within the acceptable accuracy range.

Hypothesis Formulation

The project utilizes a two-model comparison framework that will be assessed from a statistical and operational perspective. The regression models target diameter (Y) using two feature subsets: an expanded set of low-cost, high reliability observations with low-cost, medium reliability observations ($\mathbf{X}_{full}$) and a restricted set of only low-cost, high reliability observations ($\mathbf{X}_{restricted}$). While the restricted model ($\mathbf{X}_{restricted}$) is preferred due to the lower number of features and high reliability, current peer-reviewed research dictates that absolute magnitude (H) from Group 2 is mathematically required to calculate diameter (Ge, 2026). As such, the framework will determine if dropping the less reliable Group 2 characteristics results in statistically and operationally significant degradation in accuracy to predict target diameter which then justifies the observational penalties of the expanded model ($\mathbf{X}_{full}$).

The hypothesis determines if dropping the less reliable Group 2 characteristics falls within an acceptable operational threshold (δ), quantified as:

$$\Delta R^2_{test} = R^2_{Model(\mathbf{X}_{full})} - R^2_{Model(\mathbf{X}_{restricted})}$$

Statistical Hypothesis

The statistical hypothesis tests if the restrictive feature subset of Group 1 characteristics maintains a statistically acceptable degree of accuracy in determining target diameter (Y) relative to the expanded feature subset of Group 1 and Group 2 characteristics.

Statistical Null Hypothesis ($H_{0,stat}$): $\Delta R^2_{test} \leq 0$

The inclusion of Group 2 features results in no improvement or a decrease in the out-of-sample ΔR^2 on the testing dataset over the baseline model on Group 1 features only.

Statistical Alternative Hypothesis ($H_{1,stat}$): $\Delta R^2_{test} > 0$

The inclusion of Group 2 features yields a statistically significant increase in the out-of-sample ΔR^2 on the testing dataset, indicating that the expanded features materially increase the accuracy of predicting the target diameter (Y).

Evaluation Criteria: Non-parametric permutation testing or bootstrapping procedures will be utilized in evaluation to calculate the empirical p-value or confidence intervals for ΔR^2_{test} to validate statistical significance.

Operational Hypothesis

The operational hypothesis tests if the restrictive feature subset of Group 1 characteristics maintains an operationally practical acceptable degree of accuracy (δ) in determining target diameter (Y) relative to the expanded feature subset of Group 1 and Group 2 characteristics.

Operational Null Hypothesis ($H_{0,ops}$): $\Delta R^2_{test} \leq \delta$

The performance gain from adding Group 2 features is less than or equal to the operational threshold (δ). The marginal accuracy increase does not justify the added complexity and increase in number of features relative to the baseline model of Group 1 features only.

Operational Alternative Hypothesis ($H_{1,ops}$): $\Delta R^2_{test} > \delta$

The performance gain from adding Group 2 features exceeds the operational threshold (δ). The accuracy improvement materially outweighs the tracking overhead and penalty making the expanded model ($\mathbf{X}_{full}$) optimal for production.

Acquire

In order to test the two-model comparison framework, a comprehensive and reliable dataset is required. The dataset needed to contain characteristics from Group 1 and Group 2 for small-bodies. The [JPL NASA Small-Body Dataset](#) was selected because NASA is an authoritative, reliable, and primary repository of small-body data that is also open-source and available to the public allowing reproduction of testing for peer-review.

The [JPL NASA Small-Body Dataset](#) contains Group 1 and Group 2 characteristics such as absolute magnitude (H), semi-major axis (a), eccentricity (e), and inclination (i) which directly relate to the problem statement and hypothesis requirements necessary for the expanded features subset and restricted feature subset.

The [JPL NASA Small-Body Dataset](#) was queried to meet the target variable diameter (Y) to be used in out-of-sample testing. The query exclusively included asteroids with a known diameter greater than zero kilometers (0 km). This ensures a valid target variable for regression training and the calculation of the coefficient of determination (R^2). Based on these parameters, a custom CSV file was generated and downloaded manually, which is preserved in the project's GitHub repository.

Prepare

Data has been prepared for regression analysis in the following ways:

- **Target Variable Isolation (diameter):** All entries without a diameter have been removed. Diameter (Y) is the target variable and is required for the training model and to calculate the coefficient of determination (R^2) during out-of-sample testing.
- **Exclusion of High-Cost Features (albedo):** Albedo represents a Group 3 characteristic and was explicitly dropped from the dataset. Removing this feature ensures the model does not rely on cost-prohibited data and maintains operational cost-effective analysis. This also prevents data leakage between albedo and the target diameter.
- **Feature Subset Selection:** The dataset was restricted to specific characteristics required to construct the two-model comparative framework. This isolates the low-cost, high-reliability Group 1 characteristics (*e.g., semi-major axis a , eccentricity e , inclination i*) and the low-cost, medium-reliability Group 2 characteristic (*absolute magnitude, H*) needed to test the operational hypothesis.
- **Feature Completeness:** Entries with missing values in the selected feature subsets were removed. This ensures the regression algorithms receive a complete feature matrix and that any observed variance in R^2_{test} is attributable to specific feature subsets and not fluctuations in sample size or missing data.

Analyze data

- <select analytical technique, build models>

Report

- <communicate results>

Act

- <apply results, connect results with the chosen problem statement or business question>

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